Molecular dynamics simulation of Ag crystal

Davide D'Avenio

Abstract

In this project we build the code for a Molecular Dynamics simulation from scratch in Matlab. We go through a series of checks and corrections to obtain a working simulation of bulk atoms, testing everything on a FCC lattice cell of Ag atoms. Finally, we use the code to study diffusion of an addatom placed on an infinite slab of a FCC Ag crystal sliced along two different crystal planes.

Contents

1 Part A.1: Sharp Cutoff at low temperature

1.1 Theory

Our MD simulation is implemented through the velocity Verlet algorithm, which requires computing, from the interatomic potential (which we approximate as the Lenard-Jones potential), the forces on each atom. As is, computing the forces is a $O(n^2)$ operation, which wouldn't be very practical for large simulation cells. Instead, we assume that the small contributions to the force given by far away atoms can be approximated to zero. We introduce a sharp cutoff radius of $r_c = 4.5\text{Å}$ in the potential, after which we approximate it to 0, giving an operation complexity of $O(n)$. In our code, we are not actually getting this improvement, as we use the cutoff distance to rebuild the neighbors list at each iteration but the code can be further refined from this point in order to remove this limitation.

Because of this approximation, we introduce a error in the conservation of mechanical energy of the system, which grows with the size of the timestep. Thus, we have to find a timestep that is in the sweet spot between being large enough that we are able to simulate an interesting amount of time (with sensible computation times) and small enough that the energy is being conserved to an acceptable degree.

1.2 Timestep optimization at 20K

We take a set of 6 possible timesteps: $dt = [5, 10, 15, 20, 25, 30] fs$ and we run a 5000 steps (after thermalization) simulation for each one at a starting temperature of 20K. To choose the best timestep, we monitor energy fluctuation over the whole simulation and we choose the timestep that gives a value of $\frac{\delta E}{<E>}$ closest but smaller then 10^{-5}.

In figure 1 we can see that the best timestep comes out to be $dt = 25fs$. Figure 2 shows how energy and temperature oscillate as the simulation progresses. It is notable to see that temperature oscillates around a value that is higher than $T_{in}/2$. This is because we are using a finite sized crystal starting from the ideal Bravais lattice configuration, which is not its real minimum energy configuration.

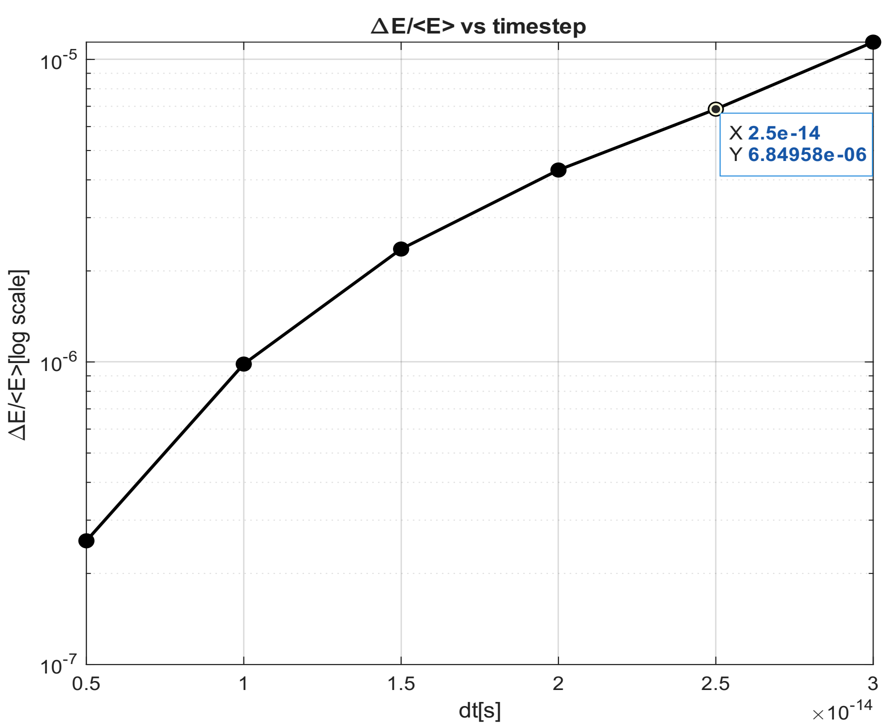


Figure 1: Relative energy fluctuations plotted against possible timesteps, chosen timestep is highlighted

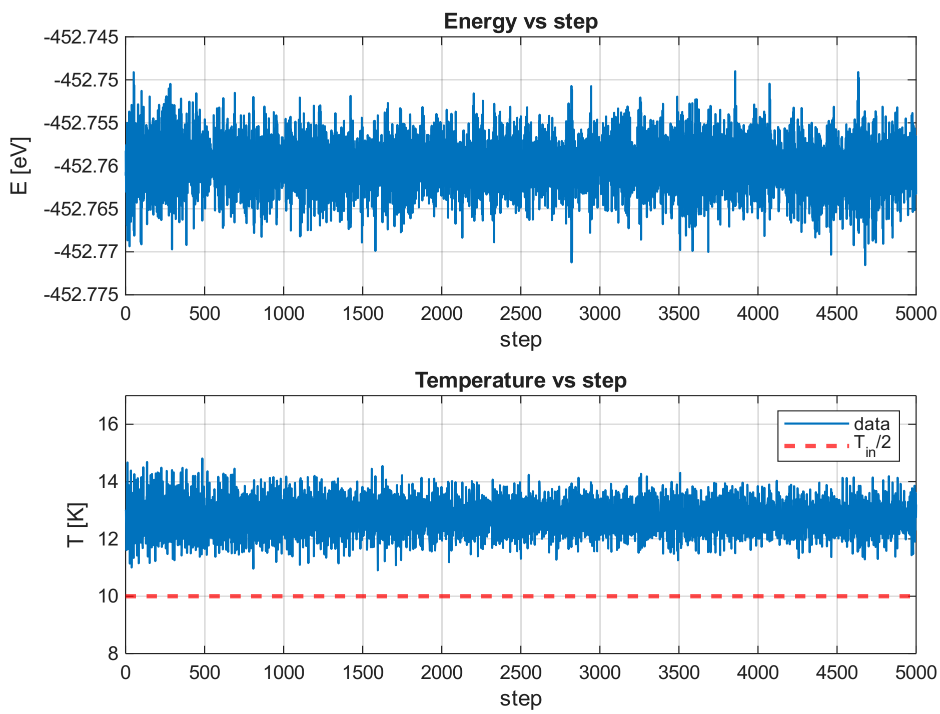


Figure 2: Behaviour of temperature and energy as the simulation progresses. The dotted line in red represents the ideal $T_{in}/2$ where temperature should stabilize if we started from the real minimum energy configuration

2 Part A.2: failure of Sharp Cutoff at higher temperature

2.1 Theory

As we approach higher energies, the sharp cutoff approach starts showing its weakness. The problem we face is that we are introducing a discontinuity in the forces by using a potential that is not derivable. To solve this, we introduce a second radius, r_{poly}, which will define an area where we substitute the L-J potential with a polynomial junction between it and $U = 0$. By using this method with a 7th order polynomial, we restore the continuity of the derivatives of the potential up to 3rd order.

2.2 Energy conservation failure at 600K

We run 5000 steps simulations for timesteps in a range from $10^{-16}s$ to $10^{-14}s$, setting the initial temperature parameter to $T_{in} = 600K$.

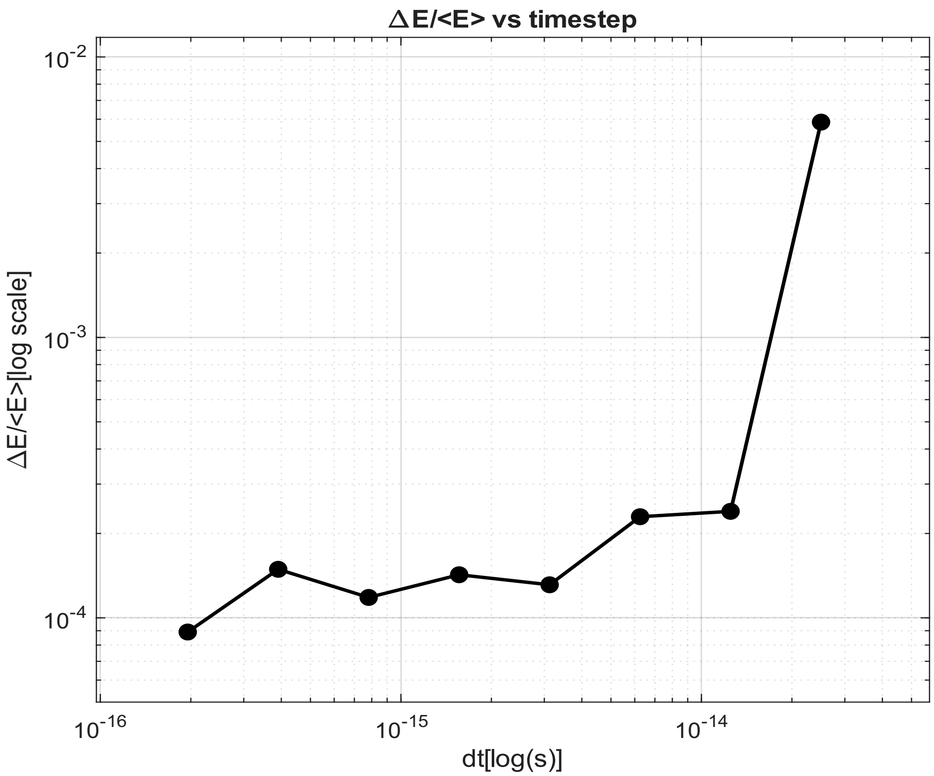
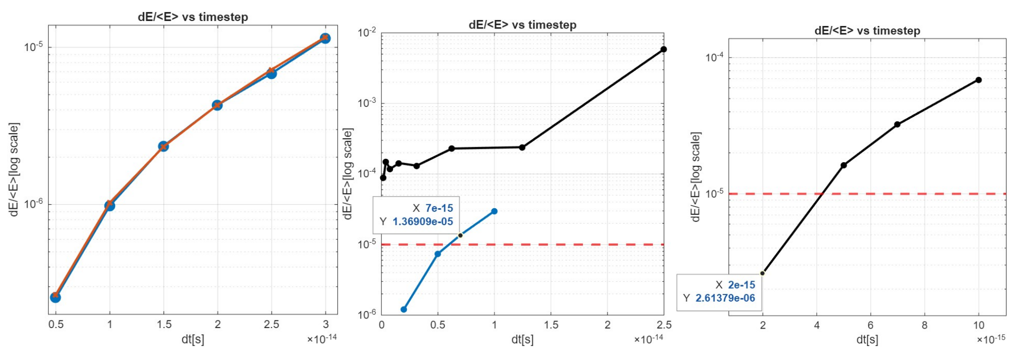


Figure 3: Energy fluctuations at different timesteps at 600K. We can see that energy conservation does not reach the threshold of 10^{-5} using the sharp cutoff approach.

We can see in image 3 that no timestep clears the threshold of $\frac{\delta E}{<E>} < 10^{-5}$ as expected. By going to even smaller timesteps, maybe $10^{-17}s$, one could argue that we could eventually clear the energy conservation threshold. However, by employing timesteps of such small scale, it would become basically impossible to simulate meaningful timescales in a finite amount of computation time.

3 Part A.3: Polynomial Junction

Seeing the failure of sharp cutoff in section 2.2, we proceed with the implementation of the polynomial scheme outlined in section 2.1. We run the adjusted code at $T_{in} = 20K$ to check consistency with the sharp cutoff solution, results in figure 4 (a). Then we check the set $[2, 5, 7, 10]fs$ for the optimal timestep at $T_{in} = 600K$ and at $T_{in} = 1200K$. We obtain that the best timesteps are, respectively, $7fs$ and $2fs$, as shown in figure 4 (b) and (c).



(a) (b) (c)

Figure 4: (a) Energy fluctuations vs timestep comparison between sharp cutoff approach (blue) and polynomial junction (red) at $T_{in} = 20K$. (b) Optimal timestep search for $T_{in} = 600K$ (blue). The highlighted point is the chosen timestep, which is just slightly above threshold (red dashed line). The black line shows the previous results from the sharp cutoff approach, clearly marking the improvement of the new method. (c) Optimal timestep search for $T_{in} = 1200K$.

4 Part B.4: Drift of the crystal cell

4.1 Theory

Now that we have an approach to potential energy computation that is stable enough to simulate a wide temperature range, we can address a new problem that becomes more evident at higher temperatures.

The simulated crystal cell presents two collective motion features: drift of the center of mass (CoM) and angular motion around it. Both of these effects stem from finiteness of the cell; since we are extracting starting velocities randomly from a distribution, at finite atom numbers we will have a sample average of the velocities that is slighly off the distribution mean.

We will fix the two motion modes independently. We fix the drift by renormalizing the extracted velocities with their sample average, resulting in the formula:

$$v_i \to v_i' = v_i - \frac{1}{N}\sum_i^N v_i \tag{1}$$

In section 6.1, we will see how rotation gets fixed naturally when we implement periodic boundary conditions.

4.2 Comparison of drift and no drift simulations at 1200K

To check whether the renormalization procedure outlined in 4.1 works correctly, we run a long ($T = 25ps$) simulation with $T_{in} = 1200K$ twice; once without renormalization and once with it. Figure 5 shows the displacement of the CoM in the two cases.

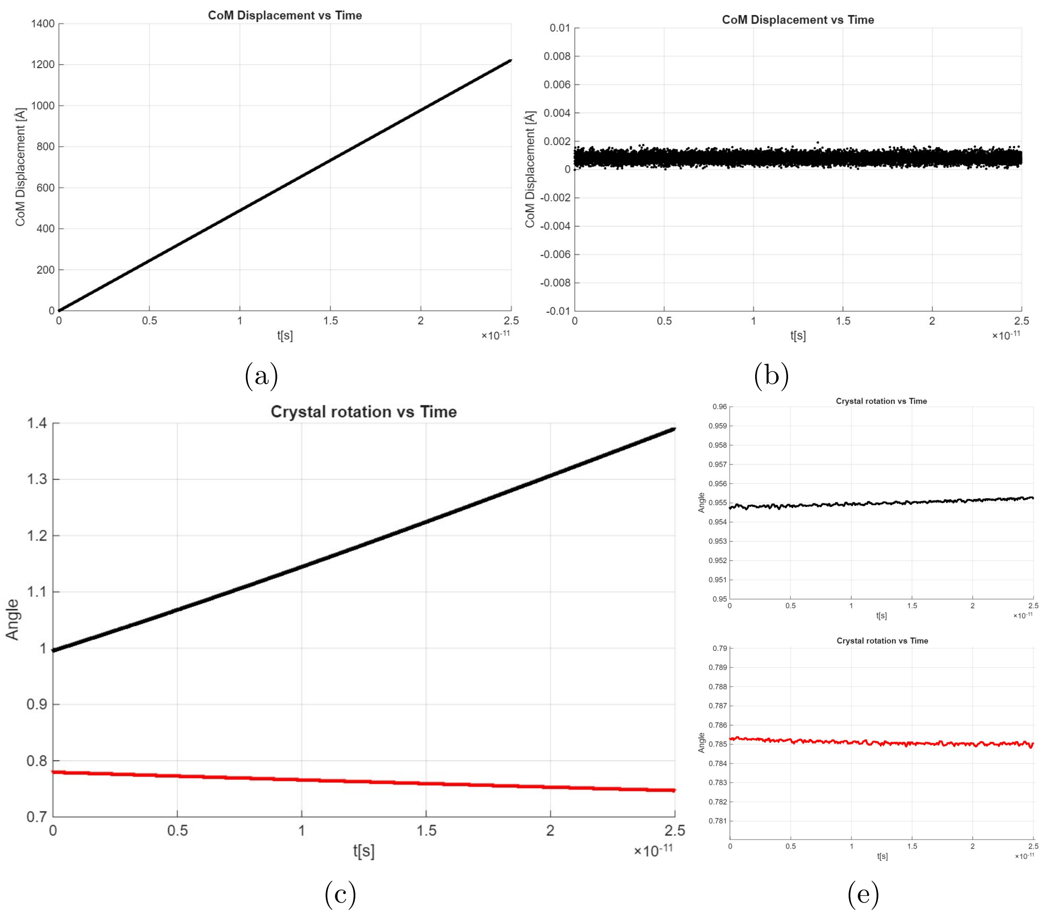


Figure 5: (a) CoM displacement at 1200K without drift correction and (b) with drift correction. CoM displacement gets fully eliminated. (c) Crystal rotation computed from the position of atom number 1 of the configuration with respect to the center of mass. The black line is $\phi = \cos^{-1}(z/r)$ and the red line is $\theta = \tan^{-1}(y/x)$, with x,y,z distances from the CoM along the axes and $r^2 = x^2 + y^2 + z^2$. (d) Crystal rotation after applying the drift correction. We can see it greatly mitigated but not eliminated completely. PBC will be needed to remove it.

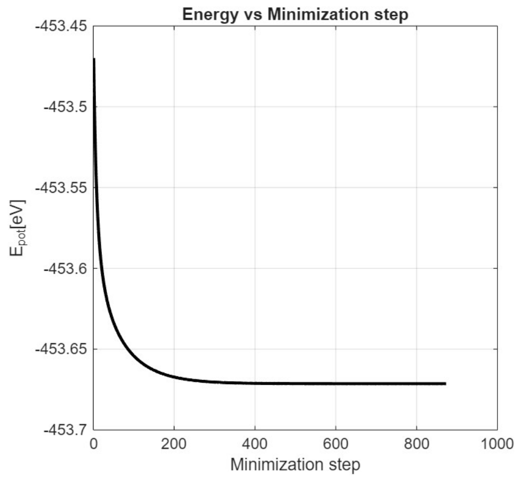
5 Part B.5: Steepest Descent

5.1 Theory

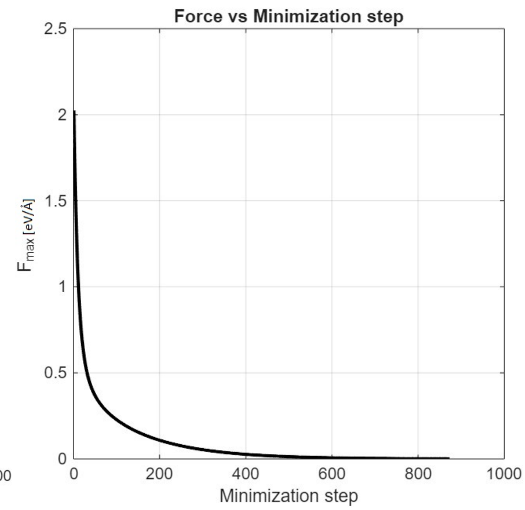
The finiteness of the crystal cell we are using takes with it another divergent property from the infinite crystal case: the regular lattice arrangement we are currently giving as input to the simulation is not the minimum energy configuration. This causes the mismatch between expected final temperature and the one actually measured, as we mentioned in section 1.2. The regular lattice configuration gives the correct minimum energy configuration for bulk atoms, however in our 256-atoms cell we have a highly non-negligible contribution from surface and close-to-surface atoms. To account for this feature, we implement a steepest-descent algorithm that is to be applied to the initial configuration before starting the simulation. The algorithm follows the gradient of the potential energy, moving the atoms to more stable configurations until the computed forces fall under a tolerance parameter, which we set at $\epsilon = 0.001 eV/\text{Å}$

5.2 Simulations starting from Minimized Configurations

We apply the steepest descent algorithm to the 256-atom cluster, monitoring the lattice energy as shown in figure 6.



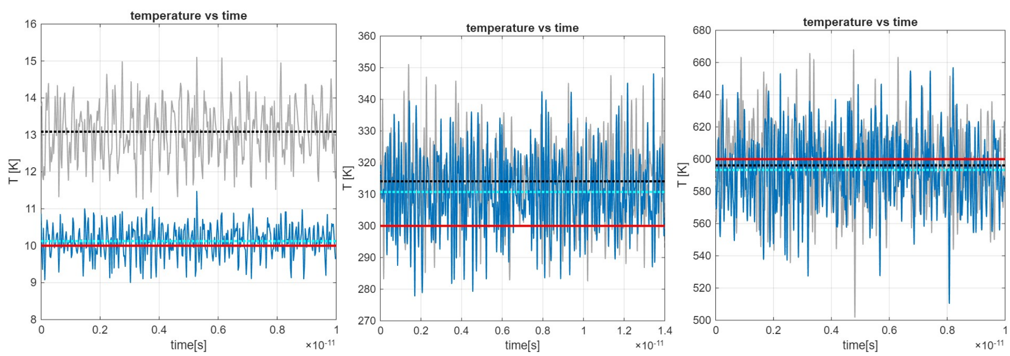
(a)



(b)

Figure 6: (a) Energy minimization process for the 256-atoms cell. As expected, the curve is monotonically decreasing, as the algorithm follows the energy gradient downwards. (b) Max force acting on the atoms as a function of the minimization steps. Again, we see a monotonically decreasing behavior; however this is not guaranteed and depends on the seed. The steepest descent algorithm has no problems even if the maximum force increases at some steps since it uses energy as a guide towards the stable configuration.

With the new minimized configuration we run $T = 10ps$ simulations for the previously explored temperatures $T_{in} = [20, 600, 1200]K$. Figure 7 shows how temperature fluctuation match the expected final temperature for the three different cases.



(a) (b) (c)

Figure 7: (a),(b),(c) Temperatures (solid grey line: without configuration optimization. Solid blue line: with optimization) and their average (black dotted line: without configuration optimization. Dotted light blue line: with optimization) as a function of time for the cases $T_{in} = 20K$ (a), $T_{in} = 600K$ (b) and $T_{in} = 1200K$ (c). The solid red line shows in each case the target temperature of $T_{target} = T_{in}/2$. Optimizing the atom configuration decreases T_{avg} in all three cases. In (c) however we can see how T_{avg} is already below the target temperature and the optimization procedure actually brings the simulation farther from it. This is due to the temperature being too high for the harmonic approximation of the potential and thus equipartition principle failing to divide evenly then energy between potential and kinetic (more goes to potential).

6 Part C.6: Periodic Boundary Conditions

6.1 Theory

In order to reach a better approximation of an infinite crystal, since we are interested in simulating the bulk behavior of the lattice, we can implement periodic boundary conditions(PBC). We apply the following correction to all computation of distances and neighbors:

$$x^{pbc} = x - floor(\frac{x}{L_x}) \tag{2}$$

Where L_x is size the periodicity for the boundary condition (along the x direction in this case, but in 3D we would have L_x, L_y, L_z). This results in our simulation cell being replicated infinitely many times in each direction, thus correcting the energy of surface and close to surface atoms, which now would "see" the amount of neighbors they would have in bulk. Another effect of this correction is that it removes the angular motion of the cell, since the new replica of the crystal balance out the forces. This approach is very powerful but presents some limitations: (a) beyond L_i the atoms behave the exact same as the ones in the original cell, thus giving a completely correlated motion, (b) if the neighbor cutoff radius is too big compared to L_i, atoms will self interact (the mathematical condition to avid this is $L_{min} = 2r_c$ (c) long range effects are badly reproduced if the simulation cell is not very large. Any defect (vacancies, inclusions...) introduced will also be present in all replicas, making it impossible to simulate an isolated one.

6.2 Constitency test of PBC Corrected Crystal

To check that the PBC correction is working properly with reliable results, we run 2, $T = 10ps$ long simulations at $T_{in} = 20K$ and $T_{in} = 1200K$. For both of them, we monitor that the average energy fluctuations don't go beyond our threshold of 10^{-5} and we also compare average energy and temperature with the isolated cluster case.

Temperature	$\Delta E/\langle E \rangle$	$\Delta T/\langle T \rangle$
20K (Cluster)	$7.23 \cdot 10^{-6}$	0.404
20K (PBC)	$1.658 \cdot 10^{-5}$	0.956
1200K (Cluster)	$2.614 \cdot 10^{-6}$	0.0375
1200K (PBC)	$2.316 \cdot 10^{-6}$	0.0362

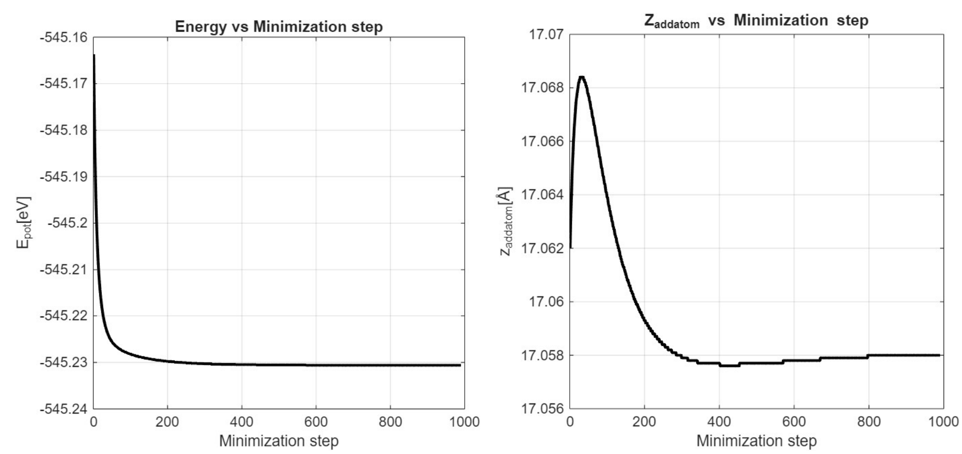
Table 1: Energy and temperature fluctuations before and after applying PBC. In the 20K PBC case we get a value that is sightly above threshold but we keep it as it does not go beyond it much.

7 Part C.7: Addatom minimization

Now that we have a working bulk simulation, we can use it explore the behavior of an addatom placed above an infinite slab of our crystal.

We remove the periodic boundary condition on one of the directions, namely z, and we create a new input configuration, adding a 257th atom 2.5Å above the cell in the z direction, centered in the middle of the cell in the x and y directions. In this way, we obtain a crystal slab sliced along the (100) plane plus the addatom on top.

To obtain the minimum energy configuration, we put the new 257 atom cluster through the steepest descent algorithm. Figure 8 shows the behavior of the energy and the z position of the addatom as the system approaches the minimum energy configuration.



(a) (b)

Figure 8: (a) lattice energy vs steepest descent step for the addatom configuration (b) addatom z position vs steepest descent step

8 Part C.8: Addatom diffusion

We run a $T = 45ps$ simulation at $T_{in} = 1750K$ with a timestep of $\Delta t = 3fs$, using the minimized configuration from section 7. We monitor the energy fluctuations and the energy, and we trace the addatom trajectory over time. Figures 9 and 10 show the results.

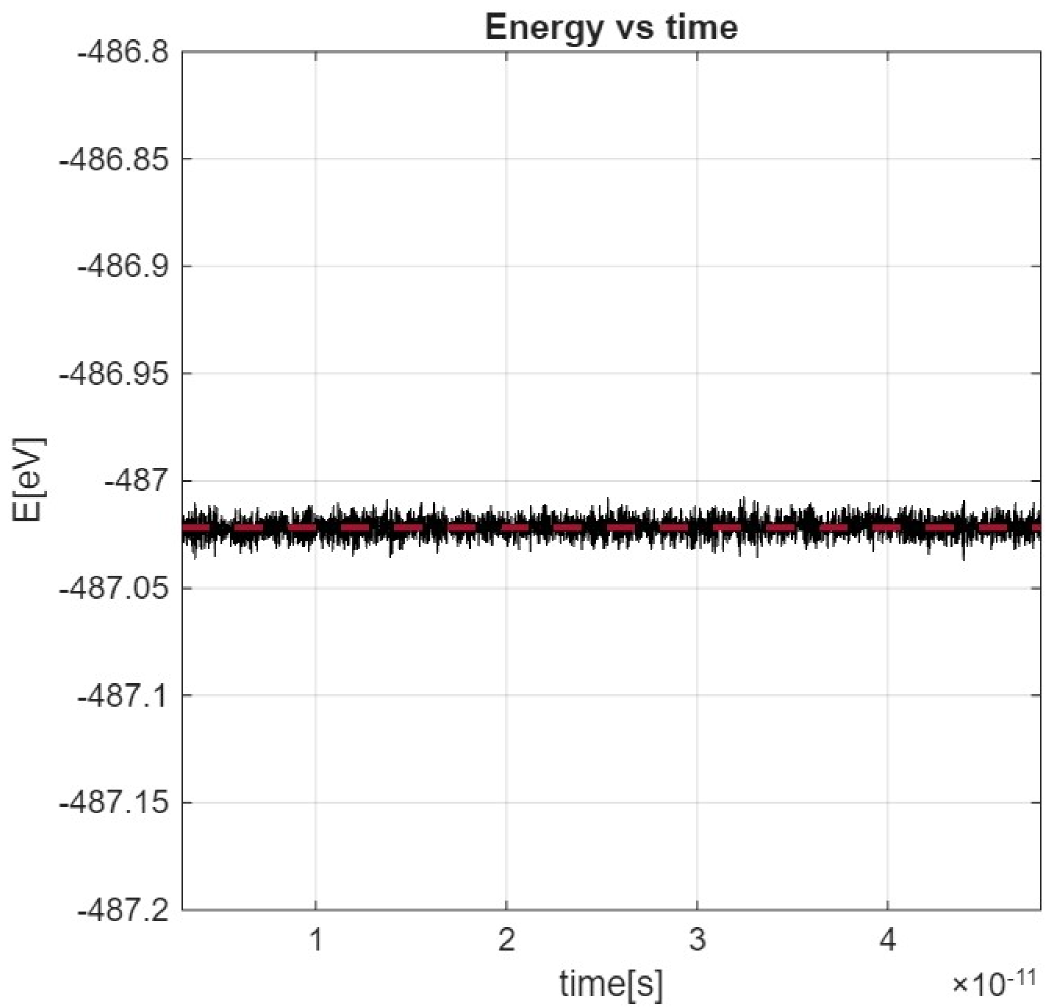


Figure 9: Energy vs timestep, red dashed line shows the average energy. Energy fluctuations stay below threshold, with their average being $\Delta E/\langle E\rangle = 8.3531 \cdot 10^{-6}$

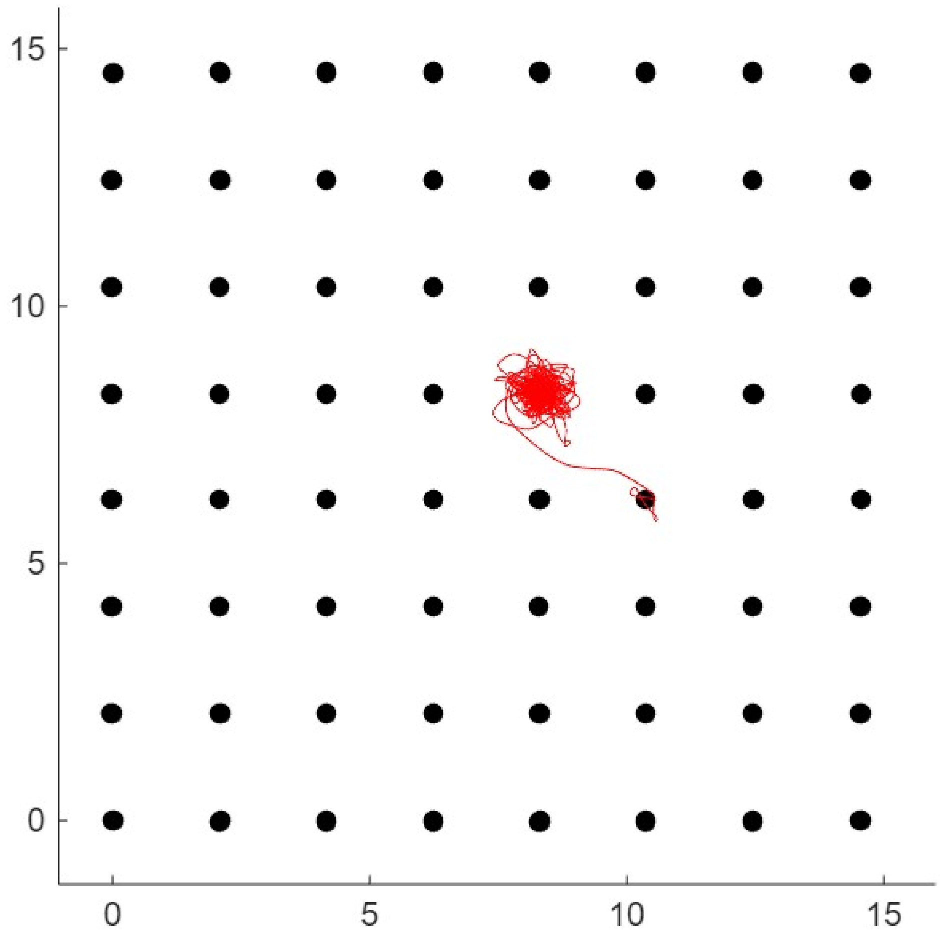


Figure 10: Diffusion trajectory of addatom on the (100) plane.

We repeat the experiment with a different configuration, this time sliced along the (111) direction, pretreated with steepest descent as before. Figures 11 and 12 show the outcome.

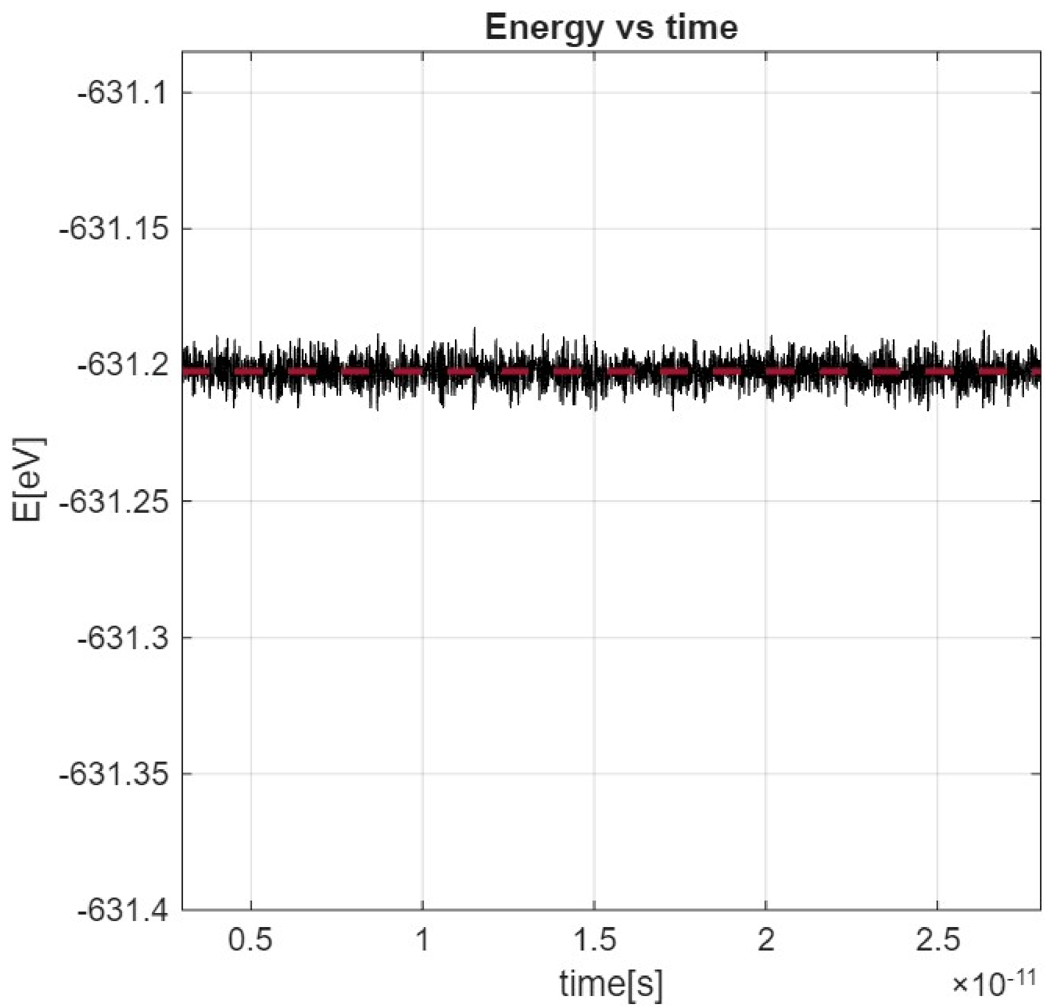


Figure 11: energy vs timestep, red dashed line shows the average energy. Energy fluctuations stay below threshold, with their average being $\Delta E/\langle E\rangle = 7.4761 \cdot 10^{-6}$

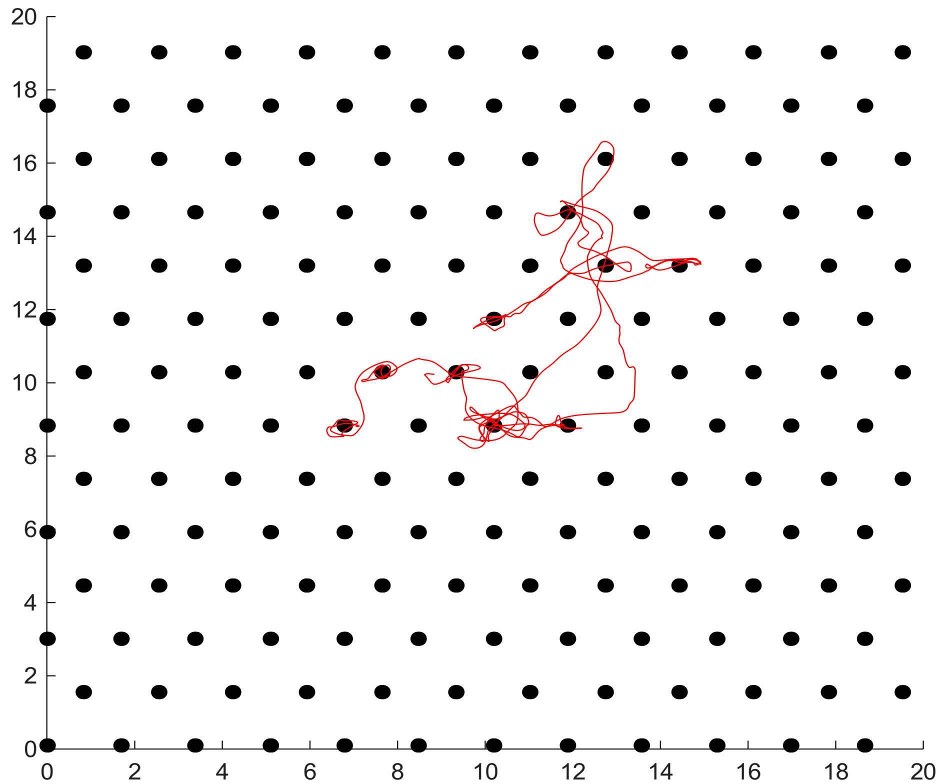


Figure 12: Diffusion trajectory of addatom on the (111) plane.

Comparing the two trajectories, two differences emerge. The sites where the addatom resides are different and the frequency of the jumps is higher in the (111) case, leading to a shorter diffusion time. This could be explained by a lower binding energy reached by the addatom in the sites available in that configuration, making it easier for a thermal fluctuation to produce a jump.